

Note: This project was generated by Manus, an autonomous AI agent, based on the user's requirements.

SRE Bot - Enterprise-Ready AI Agentic Bot

python 3.11+

FastAPI 0.109.0

LangGraph 0.0.26

Kubernetes v1.29

An enterprise-ready AI Agentic Bot designed for Site Reliability Engineering (SRE) tasks. This bot provides a conversational interface to manage and troubleshoot complex cloud-native environments, with a strong focus on security, data residency, and multi-LLM support.

Table of Contents

- [Features](#)
- [Architecture](#)
- [Getting Started](#)
 - [Prerequisites](#)
 - [Installation](#)
 - [Configuration](#)
- [Usage](#)
 - [API Server](#)
 - [Command-Line Interface \(CLI\)](#)
 - [Interactive Mode](#)
- [Tools](#)
- [Deployment](#)
 - [Docker Compose](#)
 - [Kubernetes](#)
- [Testing](#)

- [Roadmap](#)

Features

- **Multi-LLM Support:** Seamlessly switch between different LLM providers:
 - **OpenAI:** GPT-4, GPT-3.5
 - **Anthropic:** Claude 3.5 Sonnet, Claude 3 Opus
 - **Ollama:** Support for local models like Llama 3, Mistral, etc.
- **Agentic Framework:** Advanced reasoning, planning, and tool execution capabilities using LangGraph.
- **SRE Tool Suite:** A rich set of tools for:
 - **Kubernetes:** Pod/deployment management, log inspection, resource description.
 - **Observability:** Prometheus queries, health checks, log analysis.
 - **Network Diagnostics:** Connectivity tests, DNS lookups, latency measurement.
- **Enterprise-Grade Security:**
 - **Read-Only by Default:** Strict read-only permissions for cluster and cloud resources.
 - **Approval Workflows:** Human-in-the-loop for sensitive operations.
 - **Data Residency:** Support for local LLMs (Ollama) to keep data within your perimeter.
 - **RBAC:** Fine-grained access control for Kubernetes and cloud resources.
- **Multiple Consumption Modes:**
 - **REST API:** A FastAPI-based server for programmatic integration.
 - **Command-Line Interface (CLI):** An intuitive CLI for direct interaction.
 - **Interactive Chat:** A REPL-style interface for conversational workflows.
 - **MCP Server (Future):** Integration with Model Context Protocol.
- **Cloud-Native Deployment:** Ready-to-use Docker and Kubernetes configurations.

Architecture

The SRE Bot is built on a modular, microservices-friendly architecture.

Core Components

1. **API Server (FastAPI):** The main entry point for all interactions. It exposes REST endpoints for chat, tool execution, and management.
2. **Agent Engine (LangGraph):** The brain of the bot. It orchestrates the entire workflow, from understanding user intent to planning, executing tools, and generating responses.
3. **LLM Integration Layer:** An abstraction layer that provides a unified interface for multiple LLM providers, handling provider-specific logic and API calls.
4. **Tool Registry:** A central repository for all available tools. It allows for dynamic registration and execution of tools based on the agent's plan.
5. **SRE Tools:** A collection of specialized tools that interact with external systems like Kubernetes, Prometheus, and cloud provider APIs.
6. **Configuration Module:** Manages all application settings, including API keys, model names, and security policies.

For a more detailed breakdown, see the [Architecture Documentation](#).

Getting Started

Prerequisites

- Python 3.11+
- Docker and Docker Compose (for local deployment)
- `kubecttl` (for Kubernetes deployment)
- Access to an LLM provider (OpenAI, Anthropic) or a running Ollama instance.

Installation

1. **Clone the repository:**
2. **Install dependencies:**

Configuration

1. **Create a `.env` file** from the example:
2. **Edit the `.env` file** to add your API keys and configure the desired LLM provider:

Usage

The SRE Bot can be used via the API server or the CLI.

API Server

Start the API server:

Bash

```
python -m src.api.main
```

The API will be available at `http://localhost:8000` . You can access the interactive documentation at `http://localhost:8000/docs` .

Example API Call:

Bash

```
curl -X POST http://localhost:8000/chat \
-H "Content-Type: application/json" \
-d '{
  "message": "List all pods in the default namespace"
}'
```

Command-Line Interface (CLI)

The CLI provides a convenient way to interact with the bot from your terminal.

Check available commands:

Bash

```
sre-bot --help
```

Single-shot command:

Bash

```
sre-bot chat "What are the deployments in the 'kube-system' namespace?"
```

Interactive Mode

For a conversational experience, use the interactive mode:

Bash

```
sre-bot chat --interactive
```

This will start a REPL where you can have an ongoing conversation with the bot.

Tools

The SRE Bot comes with a powerful set of tools to perform various SRE tasks. You can list all available tools using the CLI:

```
Bash
```

```
sre-bot tools
```

For detailed information on a specific tool:

```
Bash
```

```
sre-bot describe-tool getpods
```

For a complete list and descriptions, see the [Tools Documentation](#).

Deployment

Docker Compose

For a quick local deployment, use Docker Compose. This will start the SRE Bot API server and an optional Ollama container for local LLM inference.

1. **Ensure your `.env` file is configured.**
2. **Run Docker Compose:**

Kubernetes

A production-ready Kubernetes deployment is available in the `deployment/kubernetes` directory. The manifests include:

- A dedicated `sre-bot` namespace and service account.
- Fine-grained RBAC roles for read-only access to cluster resources.
- A Deployment with 2 replicas for high availability.
- A Service and Ingress for exposing the bot.
- ConfigMaps and Secrets for configuration.

To deploy:

1. **Update the secrets** in `deployment/kubernetes/deployment.yaml` with your base64-encoded API keys.
2. **Apply the manifests:**

Testing

The project includes a suite of unit and integration tests.

Run all tests:

```
Bash
```

```
pytest
```

Run tests with coverage:

```
Bash
```

```
pytest --cov=src
```

Roadmap

This project follows the 10-sprint roadmap outlined in the initial requirements, covering infrastructure, platform, observability, and advanced features.

- **Release 1 (In Progress):** Infrastructure & Platform Foundations
- **Release 2:** EKS Add-ons & Observability
- **Release 3:** Network Troubleshooting & Automation
- **Release 4:** Migration Features & Guardrails
- **Release 5:** Security Hardening & Enablement

For more details, refer to the original `SRE_Bot_Roadmap.md` document.